

determined by certain values of m (fermion mass) and g (coupling constant) [66]. In this study, we utilize the variational quantum self-organizing map to distinguish between the two distinct phases characterized by this phase transition.

The problem of classifying the two distinct phases of the Schwinger model using quantum machine learning techniques has been previously investigated by Kühn et al. [58]. However, the primary distinction between our methodology and theirs is the implementation of an unsupervised machine learning strategy in our approach, which does not necessitate the presence of labels. Nevertheless, our framework for data generation closely follows the one described in [58]. We describe the process of data generation below. The Lagrangian of the Schwinger model is given by:

$$\mathcal{L} = \psi(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{g\theta}{4\pi}\epsilon_{\mu\nu}F^{\mu\nu}, \quad (20)$$

where ψ corresponds to the fermion field, $D_\mu = \partial_\mu + igA_\mu$ is the covariant derivative, and A_μ corresponds to the gauge field. The coupling constant g governs the strength of the interaction between the fermions and gauge bosons, m is the fermion mass, and $F_{\mu\nu}$ corresponds to the electromagnetic tensor. The lattice implementation of the Schwinger model can be achieved by employing the staggered fermion techniques [67]. After the Jordan-Wigner transformation the Hamiltonian of the Schwinger model is given by

$$H = J \sum_{n=0}^{N_s-2} \left(\sum_{i=0}^n \frac{Z + (-1)^i}{2} + \frac{\theta}{2\pi} \right)^2 + \frac{w}{2} \sum_{n=0}^{N_2-2} [X_n X_{n+1} + Y_n Y_{n+1}] + \frac{m}{2} \sum_{n=0}^{N_2-1} (-1)^n Z_n. \quad (21)$$

The Hamiltonian (with $N_s = 4$, i.e. 4 spatial lattice sites) is diagonalized in order to derive the lowest eigenstates corresponding to a specific value of the ratio m/g . The eigenstates are then labeled based on the order parameter that governs the phase transition of the Schwinger model, i.e. the expectation value of an averaged electric field

$$E = \frac{1}{N} \sum_{n=0}^{N-1} \sum_{i=0}^{N-1} \frac{Z_i + (-1)^i}{2}. \quad (22)$$

If $\langle E \rangle = 0$, then the corresponding state is labeled as class ‘0’. Conversely, if $\langle E \rangle > 0$, it is denoted as the class ‘1’. It is crucial to emphasize that, similar to the unsupervised learning of the Iris dataset previously described, the labels are not included as input to the algorithm; instead, they are provided for the purpose of comparison.

We observe that the variational quantum self-organizing map (variational QSOM) not only performs classification of distinct phases, but also maintains the

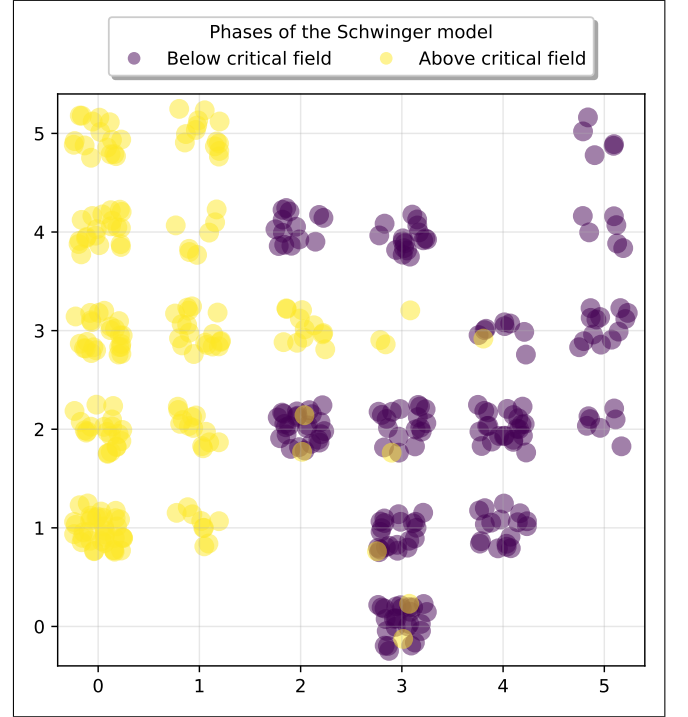


FIG. 6. The variational quantum self-organizing map (variational QSOM) of the Schwinger model not only performs classification of distinct phases, but also maintains the topological structure of the Hilbert space in the reduced-dimensional representation, that is the map maintains the topological characteristics of the original data by grouping together the quantum states from the same phase. The legends ‘Below critical field’ and ‘Above critical field’ refer to states with $\langle E \rangle = 0$ and $\langle E \rangle > 0$, respectively.

topological structure of the Hilbert space in the reduced-dimensional representation, that is, the map maintains the topological characteristics of the original data by grouping together the quantum states from the same phase, as shown in Fig. 6.

V. DISCUSSION

In this article, we have proposed a novel quantum neural network architecture for unsupervised machine learning of classical and quantum data on a near-term quantum computer. The architecture underlying the proposed quantum neural network is inspired by and based on Kohonen’s *self-organizing map* (SOM). The proposed algorithm scales linearly in the number of quantum data samples provided that there exists a predefined number of quantum states parametrized by the weight vectors in the output layer of SOM. We have demonstrated the unsupervised learning capability of the proposed quantum algorithm by creating low-dimensional representation maps for both classical and quantum datasets. The algorithm generates lower-dimensional maps that preserve the topo-